

5. There are many sex-linked genetic diseases in humans which cause changes in phenotype. X-linked inheritance means that the gene is located on the X chromosome. X-linked severe combined immunodeficiency (X-SCID) is a recessive immunodeficiency disorder in which the body produces very few lymphocytes.

- (a) (i) I. Use the symbols X^D for unaffected by X-SCID and X^d for affected by X-SCID to draw a genetic diagram to show how parents who are both unaffected can produce a son with X-SCID. [2]

Parental phenotype unaffected male × unaffected female

Parental genotypes

Gametes

Space for genetic diagram

- II. **Complete the table** below linking the **four** different genotypes from your genetic diagram with their correct phenotype. [2]

Offspring genotype	Offspring phenotype
.....	
.....	
.....	
.....	

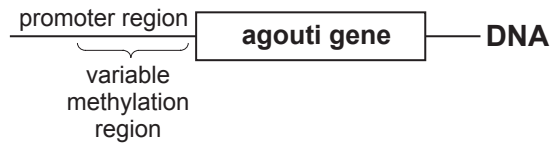
- (ii) The estimated UK incidence of X-SCID is 1 in 48 933. The population of Wales in 2018 was 3 139 000. Calculate the number of cases of X-SCID that would have been expected in Wales in 2018. **Give your answer to an appropriate number of significant figures.** [2]

Number of cases =



There are other mechanisms, such as epigenetic changes, which can alter the phenotype of organisms. One example is DNA methylation, where a methylated promoter region decreases the expression of a gene and demethylation increases the expression of a gene. This is shown in **Image 5.1**.

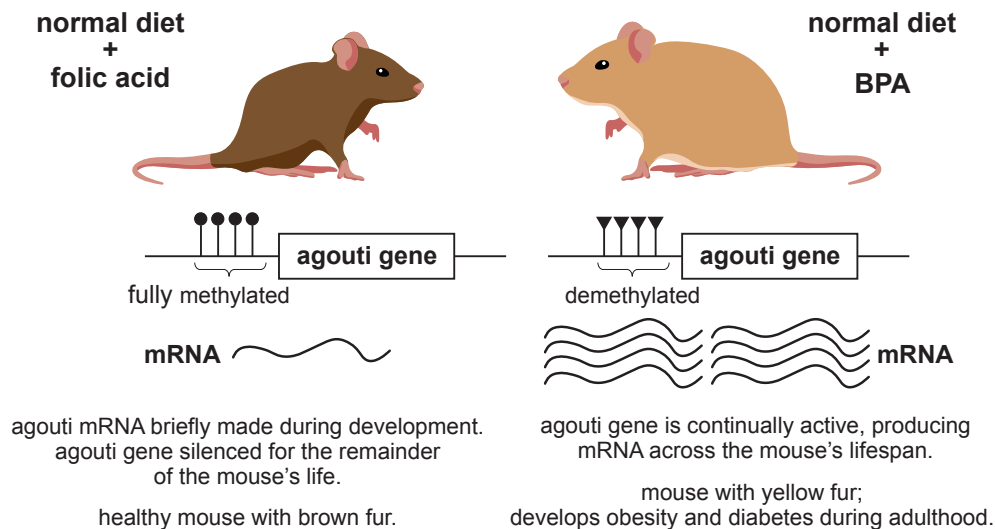
Image 5.1



Epigenetic control mechanisms can be seen in coat colour in mice which is influenced by the nutrition of the pregnant female.

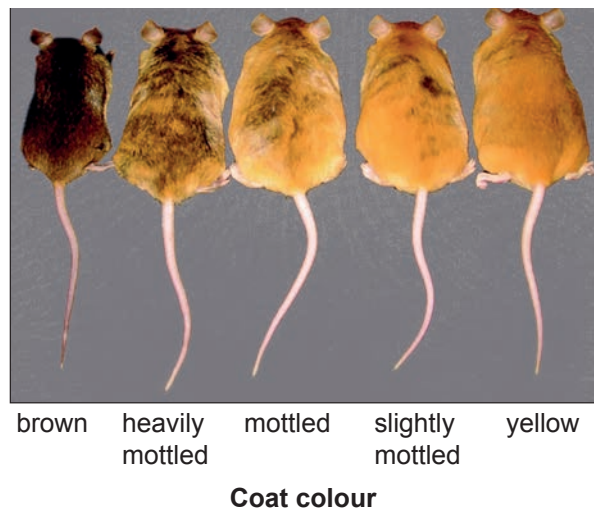
When the pregnant females are fed compounds such as BPA (used in plastic production) with their normal diet, this can cause more yellow offspring to be born. When the pregnant females' normal diets are supplemented with high folic acid, more brown offspring are born. This is summarised in **Image 5.2**.

Image 5.2

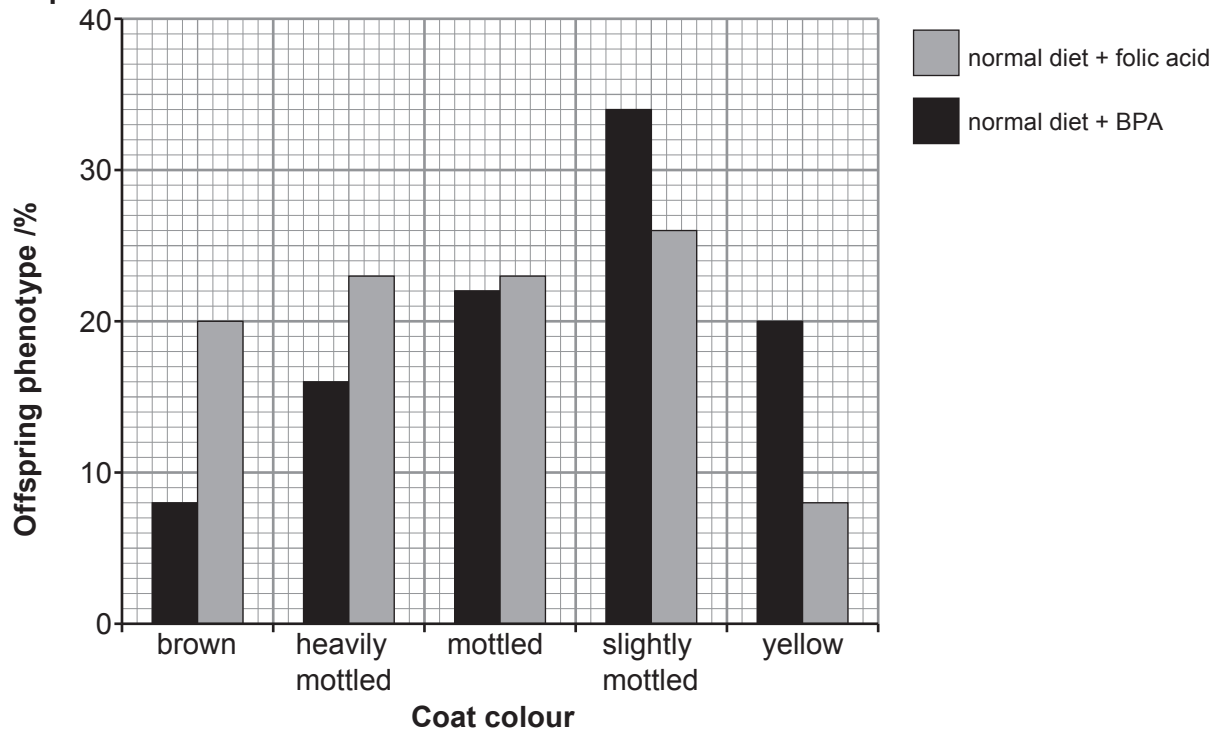


When mice were fed the two different diets, the following results were obtained, with ranges in coat colour as shown in **Image 5.3** and number of each type shown in **Graph 5.4**.

Image 5.3



Graph 5.4



- (b) (i) Using **Images 5.1, 5.2, 5.3** and **Graph 5.4**, explain the effect of the **two** different supplements on gene expression and coat colour in mice. [4]

.....

.....

.....

.....

.....

.....

.....

.....

- (ii) Use all of the information to suggest how the results of this study might influence the use of BPA in the production of food containers and packaging. [2]

.....

.....

.....

.....

